

NeuroFleetX – Frontend Documentation (User & Driver Modules)

Introduction

NeuroFleetX is an AI-powered taxi fleet management system. This document explains the User and Driver frontend modules for assessment and mentor review. The frontend is built using React.js with role-based access control.

Frontend Technology Stack

React.js, React Router DOM, Context API, REST APIs, Protected Routes.

User Module

Users log in and are redirected to the User Dashboard based on `ROLE_USER`. The dashboard provides access to New Ride Request, Ride History, Analytics, and Profile pages.

New Ride Request

Users enter origin-destination details, traffic and weather data. The system creates a ride, predicts ETA using AI services, and assigns the best available taxi.

User Analytics

Analytics shows summarized insights such as total rides, completed rides, and average ETA using aggregated backend data.

Driver Module

Drivers log in with `ROLE_DRIVER` and access the Driver Dashboard showing assigned rides, taxi status, and earnings.

Taxi Management & Rides

Drivers can view taxi information, manage ride status, and track assigned rides in real time.

Security

Role-based protected routing ensures users and drivers can only access authorized pages.

Conclusion

The frontend design ensures secure, scalable, and real-world ride-hailing functionality through clear role separation and real-time API integration.